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Department of Electronics and Communication Engineering

DEEP LEARNING APPROACH FOR CLASSIFICATION OF EYE DISEASES BASED ON COLOR FUNDUS IMAGES USING MATLAB

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ABSTRACT

- ▶ Eye disease recognition is a challenging task, which usually requires years of medical experience.
- ▶ In this work, we conducted research that can be a start for the most versatile solution.
- ▶ We tried to solve the problem of the prediction of different eye diseases using neural networks.
- ▶ The first step of this work consists of gathering all publicly available eye disease datasets and pre-processing them to make the experiments as generalized as possible.
- ▶ This application is developed using image processing, deep learning approaches.
- ▶ For extract the feature densenet architecture is used and detect the disease by using transfer learning.

LITERATURE SURVEY

S.NO	TITLE OF THE PAPER	AUTHOR	METHODOLOGY	YEAR
1	Detection of Retinal Disease Using Image Processing	Sudeepa Deshmukh, Sandeep Mishra	In this, preprocessing technique leads to enhance the boundaries and feature extraction process and along with conversion of image type and then by combining the image processing part with the machine learning part we are able to design the algorithm.	2021
2	Fully automated detection of retinal disorders by image-based deep learning	Yan Tong	A total of 36,231 fundus images were labeled by 13 licensed retinal experts. A 101-layer convolutional neural network (ResNet) and a faster region-based convolutional neural network (Faster-RCNN) were trained for image classification and identification.	2020
3	Machine learning applied to retinal image processing for glaucoma detection	Daniele M.S.Barros	This is a systematic review on the main algorithms using machine learning (ML) in retinal image processing for glaucoma diagnosis and detection. ML has proven to be a significant tool for the development of computer aided technology.	2020

SOFTWARE REQUIREMENTS

► MATLAB:

MATLAB (*matrix laboratory*) is a multi-paradigm numerical computing environment and proprietary programming language developed by MathWorks. MATLAB allows matrix manipulations, plotting of functions and data, implementation of algorithms, creation of user interfaces, and interfacing with programs written in other languages, including C, C++, C#, Java, Fortran and Python.

The MATLAB application is built around the MATLAB scripting language. Common usage of the MATLAB application involves using the Command Window as an interactive mathematical shell or executing text files containing MATLAB code.

PROPOSED WORK:

- ▶ AMD is degeneration of the macula, which is a part of the retina, responsible for the sharp, central vision needed to read, drive and for face recognition. Diabetic retinopathy (DR) is a complication due to diabetes. DR leads to several abnormalities like microaneurysms, haemorrhages, exudates, cotton wool spots, venous irregularities, new vessels and macular edema.
- ▶ All these lead to vision loss or blindness, if not detected at early stage. Early detection of eye disease is better. In our proposed work we find the eye disease with help of the deep features and transfer learning method. To extract the deep features from the image we use densenet architecture and use transfer learning method for classification.

Modules :

- Get the data
- Pre processing
- Feature extraction
- Training
- Testing
- Classification

CONCLUSION:

- ▶ In conclusion, we discussed the main fields for the application of ML into ophthalmology practice. The use of ML algorithms should be seen as a tool to assist clinicians, not to replace them.
- ▶ ML could speed up some processes, reduce the workload for clinicians, and minimize diagnostic errors due to inappropriate data integration. ML is able to extract features from complex and different imaging modalities, enabling the discovery of new biomarkers to expand our current knowledge of diseases.
- ▶ This could lead to introducing into clinical practice new automatically detected diagnostic parameters or to developing new treatments for eye diseases

THANK YOU